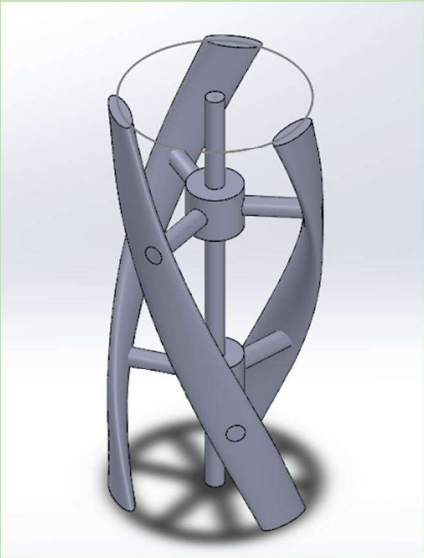
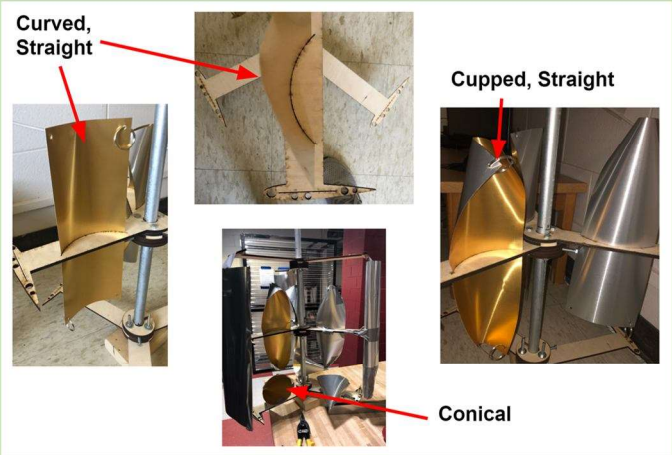
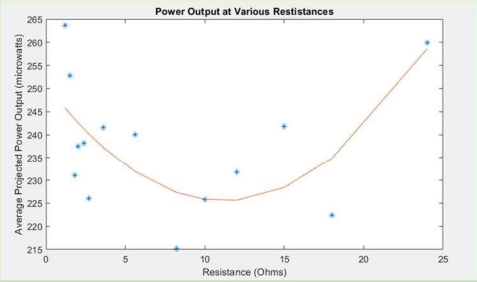
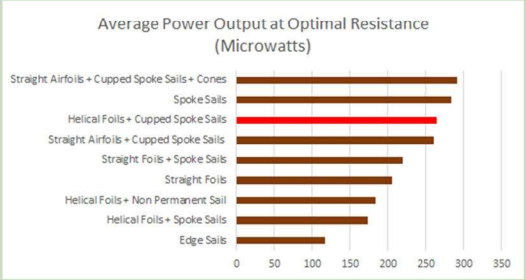
VT BREEZE

VT Breeze was an Engineering project. I worked with six other students. The 3-D rendering you see on this page was created by one of my parnters. I created the sail alternatives and led the construction of the brown airfoils. I also wrote the Matlab script that generated the output on the next page and created those graphs. The final wind turbine generated electricity and earned a 97% in Foundations of Engineering Part II at Virginia Tech in Spring 2019.





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Based on the test data, at a windspeed of 1.9 meters per second,
the optimal resistance value is 1.2 ohms, which allowed the turbine
to spin an average of 65 revolutions per minute. At this resistance,
the turbine produced an average of  0.131 volts. A standard electrical
outlet supplies 919.5 times this voltage. This resistance and voltage
would produce an average power output of 264 microwatts. 1517 times
this power would be required to run a small 4 lumen lightbulb, one
that produces as much light as four standard candles.

This iteration gathers 0.02 percent of the theoretical 1.31 watts
of windpower (Joules of total wind energy available per second)
based on the largest crossectional area of the turbine.
fx >>
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